

Evaporation

$\frac{C_v}{k_b} = \frac{d\langle E \rangle}{dk_b T} = \frac{1}{2}$ # of squared terms in the Hamiltonian with non-zero coefficients.

Each atom has a squared kinetic energy term in the Hamiltonian in x and y, giving 6 squared terms. For a system of three atoms bound together (close to equilibrium), each atom has a squared potential term in x and y. There will be three degrees of freedom that are zero: When they all move in x, when they all move in y, when they rotate together, so: $6+6-3 = 9$ d.o.f. Thus, we expect $\frac{9}{2}k_b$ slope when all atoms are bound and close to equilibrium. When the atoms get enough energy to escape the morse potential, only the 6 kinetic d.o.f. count, so here the slope is $\frac{6}{2}k_b$.

Running MD simulations of different T and evaluating their mean energy and $C_v = \frac{d\langle E \rangle}{dT}$ we get the resulting curves. At very low T, $C_v \sim \frac{9}{2}k_b$. As the temperature increases, so does the mean energy. At some point, the temperature gets so high that the energy of the molecule is large enough to evaporate it, i.e. break the bonds between the atoms. When the energy is this large, the state moves up in the morse potential, until the distance between atoms goes toward infinity. That is the steep rise we see on the left graph. This energy barrier is represented as a large peak in heat capacity. After the bonds break, the heat capacity drops to $\sim 3k_b$. It drops further than the initial state, since there are no strong bonds between the atoms (no squared terms from the potential).

